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Laminated Thin Li-Ion Batteries Using a Liquid Electrolyte

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Thin Li-ion batteries with a laminated film bag as a casing were developed by using a liquid electrolyte and a graphitized boron-doped mesophase-pitch-based carbon fiber (B-MCF) anode. The thin Li-ion batteries exhibited excellent discharge performance, long cycle life, and very low swelling under high-temperature storage. A 1.5 M solution of $LiBF_4$ in an ethylene carbonate (EC)/γ-butyrolactone (GBL) (1:3 by volume) mixed solvent is advantageous for use as the electrolyte in the laminated film bag because of its high flame point of 129°C, high boiling point of 216°C, low vapor pressure, and high conductivity of 2.1 mS/cm at −20°C. The B-MCF anode in the $LiBF_4$-EC/GBL electrolyte exhibited a high reversible capacity of 345 mAh/g, a high coulombic efficiency of 94% at the first cycle, and high rate capability. It was demonstrated that the thin Li-ion battery with a thickness of 3.6 mm has a high energy density of 172 Wh/kg, high rate capability between 0.2 and 3C rate discharge, a high capacity ratio of 50% at 1C rate discharge and −20°C, and a long cycle life of more than 500 cycles at 1C rate charge-discharge cycling. The B-MCF anode led to the high rate discharge performance and the long cycle life of the thin Li-ion batteries using the $LiBF_4$-EC/GBL electrolyte. The very low swelling and small evolution of gas under the high-temperature storage at 85°C were attributable to the stability of $LiBF_4$-EC/GBL electrolyte against the fully charged $LiCoO_2$ cathode material.
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Prismatic Li-ion batteries have been developed in order to achieve progress in terms of higher capacity, lighter weight, thinness, and safety. Recently, thin-film and prismatic polymer Li-ion batteries using polymer gel electrolytes were commercialized for some portable electronic devices. Polymer Li-ion batteries have the advantages of thinness and light weight due to the use of the laminated film bag.[1,2] However, polymer Li-ion batteries are subject to some problems in that their performance is inferior in some respects to that of Li-ion batteries. We developed a laminate-type thin Li-ion battery using a graphitized mesophase-pitch-based carbon fiber (MCF) anode whose performance and safety are superior to those of polymer Li-ion batteries and prismatic Li-ion batteries.[3] We named the thin Li-ion battery the advanced lithium-ion battery. The battery has been produced by A&T Battery Corporation for cellular phones since 2000. By using a high-viscosity organic liquid electrolyte with thermal stability, the MCF anode, and the laminated film bag, the thin Li-ion batteries with high performance and safety were fabricated as thin cells with thicknesses of 1.2 and 3.6 mm.[4] Herein, we report the cell performance and the characteristics of materials for the thin Li-ion batteries using a graphitized boron-doped mesophase-pitch-based carbon fiber (B-MCF) anode.

Experimental

The thin Li-ion batteries were constructed by using the B-MCF anode, a $LiCoO_2$ cathode, an organic liquid electrolyte, a separator, and an aluminum-plastic laminated film bag as the case. The electrolyte used for the thin Li-ion batteries was a 1.5 M solution of $LiBF_4$ in a mixture 1:3 by volume of ethylene carbonate (EC) and γ-GBL (1.5 M $LiBF_4$-EC/GBL). 1 M $LiPF_6$-EC/diethyl carbonate (DEC), 1 M $LiPF_6$-EC/methyl-ethyl carbonate (MEC), 1 M $LiPF_6$-propylene carbonate (PC), and 1 M $LiPF_6$-EC/PC electrolytes were also used for the purpose of comparison with the 1.5 M $LiBF_4$-EC/GBL electrolyte. The B-MCF was prepared by adding B_4C to MCF and highly graphitized at Petoca Corporation Limited.[5] Natural graphite prepared for Li-ion batteries (Kansai Coke and Chemicals) was also used for the purpose of comparison with the cycle performance of the B-MCF. X-ray diffraction (XRD) data of the layer spacing d_{002} and the size of the crystalline Lc along the c axis, and the surface area obtained by the Brunauer-Emmett-Teller (BET) method using N_2 gas of these carbon samples are listed in Table I. MCF and B-MCF have the advantage of low surface areas, which leads to good reversibility at the first cycle. The anode and the cathode construction were the same as those described previously.[6,7] Because the cathode, the anode, and the separator were molded before the addition of the electrolyte into the battery, the cathode and the anode were in close contact with the separator. The molding was a treatment of pressing 100 kg at 80°C on a Li-ion battery without an additive polymer. The laminated film bag, the liquid electrolytes, and the separator were produced by Syowa Aluminum, Tomiyama Pure Chemical, and Asahi Chemical, respectively. All experiments after the molding were carried out in a dry argon box. The charge-discharge characteristics were evaluated between 4.2 and 3 V at 20°C. The composition of the gas obtained from the thin Li-ion batteries after the high-temperature storage at 85°C was analyzed by gas chromatography.

Results and Discussion

Liquid electrolyte.—In the case of thin Li-ion batteries using the laminated film bag, at a high-temperature condition of more than 60°C, swelling must be small and safety is required. The electrolyte of thin Li-ion batteries must be more thermally stable than that of conventional Li-ion batteries using a metallic can. A promising electrolyte is a solution consisting of high viscosity solvents such as EC, PC, and GBL because the boiling points and flame points of the high-viscosity solvents are higher than those of the EC/DEC and the EC/MEC electrolytes used for the conventional Li-ion batteries. For the thin Li-ion batteries in the high-temperature condition, the vapor pressures of the promising electrolytes must be lower than those of the EC/DEC and the EC/MEC electrolytes. Figure 1 shows vapor pressures of 1 M $LiPF_6$-EC/DEC (1:1) electrolyte and 1.5 M $LiBF_4$-EC/GBL (1:3) electrolyte in the temperature range from 0 to 100°C. The EC/GBL electrolyte showed lower vapor pressure than the EC/DEC electrolyte. On the other hand, it is necessary to improve the discharge performance for the thin Li-ion batteries and the polymer Li-ion batteries in the low-temperature range of below 0°C.

Table I. HTT, $d_{(002)}$, Lc, and BET surface area of carbon samples.

	HTT (°C)	d_{002} (nm)	Lc (nm)	Surface area (m^2/g)
MCF	3000	0.3364	59	0.8
B-MCF	3000	0.3356	>100	0.7
Graphite	—	0.3355	>100	6

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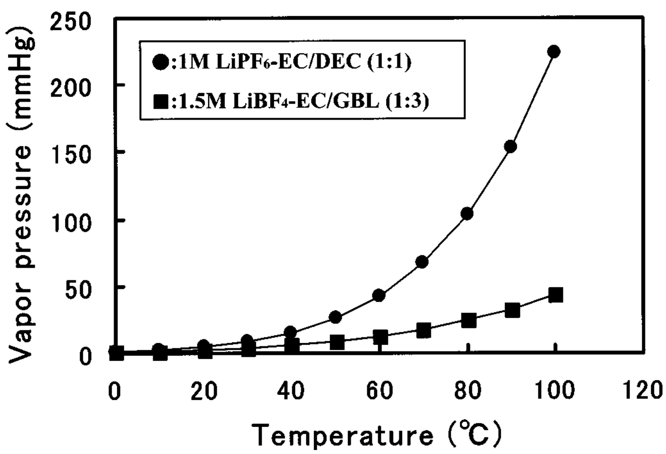

Figure 1. Dependence of vapor pressure of 1 M $LiPF_6$-EC/DEC (1:1) and 1.5 M $LiBF_4$-EC/GBL (1:3) on temperature.

Figure 2 shows the dependence of conductivity of various electrolytes on temperature. In the lower temperature range of −40 to 0°C, the conductivities of 1.5 M $LiBF_4$-EC/GBL (1:3) electrolyte were higher than those of $LiPF_6$-EC/DEC (1:1) electrolyte and $LiPF_6$-EC/PC (1:1) electrolyte. Therefore, the $LiBF_4$-EC/GBL electrolyte is advantageous for use as the electrolyte in the laminated film bag because of its high boiling point of 216°C, high flame point of 129°C, low vapor pressure of 24.6 mmHg at 80°C, and high conductivities of 6.04 mS/cm at 20°C and 2.1 mS/cm at −20°C.

B-MCF anode.—The liquid and the gel electrolytes containing PC on the graphite anode lead to a large gas evolution at the first charge.[8] The irreversible capacity of the graphite anode in the PC-based electrolyte at the first cycle is so large that the columbic efficiency becomes a low value of below 50%.[9,10] The thin Li-ion batteries using PC-based electrolytes show a large swelling because of the gas evolution. It is desirable that PC-free electrolytes be used to improve the performance and the swelling for the thin Li-ion batteries using the graphite anode. In the $LiBF_4$-EC/GBL electrolyte, even the highly graphitized B-MCF anode exhibited the high capacity of 345 mAh/g and the high reversibility of 94% at the first cycle as shown in Fig. 3. The capacity of B-MCF was 10% larger than that of MCF.[6] The efficiency in the $LiBF_4$-EC/GBL electrolyte is comparable to that in the $LiPF_6$-EC/DEC electrolyte.[6] Raman spectroscopy has indicated that highly graphitized B-MCF and B-doped graphites have a turbostratic structure at the surface.[11,12] We consider that the turbostratic structure reduces the reduction of electrolyte on the surface and produces the high reversibility.[13] Figure 4 shows variations in the discharge curves of B-MCF with discharge current densities between 50 and 1200 mA/g. The discharge capacity at 1200 mA/g showed 88% of the maximum capacity at 50 mA/g. 1200 mA/g discharge corresponds to a rate of 3.5C. Such a high rate capability of the B-MCF anode will lead to a high rate performance of thin Li-ion batteries even at 3C rate discharge. The cross section of B-MCF has a radial-like texture similar to that of MCF. The carbon fiber with the radial-like texture is advantageous in that lithium ion diffuses rapidly from the surface into the inside of the carbon fiber.[14] The carbon fiber exhibited a small resistance of interfacial reaction in comparison with that of graphite.[14] We consider that the rapid intercalation of lithium ions into graphitized carbon is essential for the further enhancement of the discharge performance and cycle life for Li-ion batteries using high viscosity solvents such as GBL, PC, and EC solvents. While the discharge curve at the high rate of 1200 mA/g showed large polarization. We consider that the large polarization may be caused by IR loss concerning the GBL-based electrolyte. Li-ion batteries using graphite anodes and the GBL-based electrolyte have never been commercialized. The cycle performance of graphite anodes in the $LiBF_4$-EC/GBL electrolyte was very poor. Lithium deposition on the surface of graphite in the EC/GBL electrolyte occurs during the high-rate charge. The GBL solvent reacts with the lithium,[15] and it may lead to the formation of a passivating film with a large resistance. However, we found that B-MCF and MCF anodes enhanced the cycle performance of the thin Li-ion batteries even in the case when the $LiBF_4$-EC/GBL electrolytes were used. The B-MCF and MCF anodes would prevent lithium from depositing on the carbon

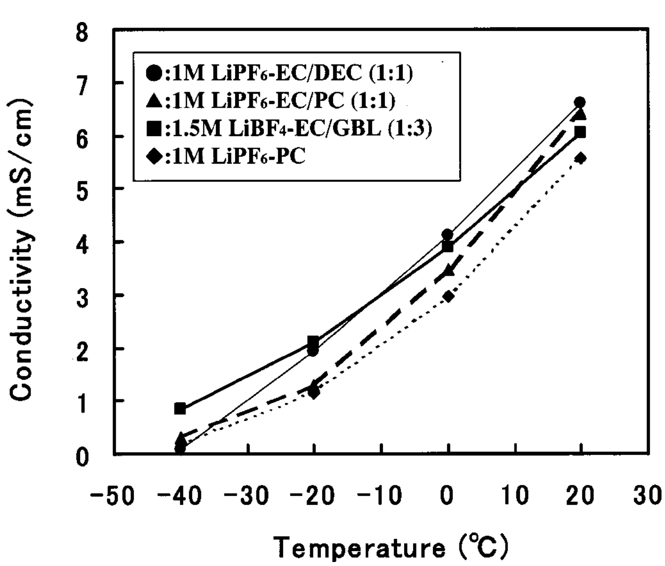

Figure 2. Dependence of conductivities of various electrolytes on temperature.

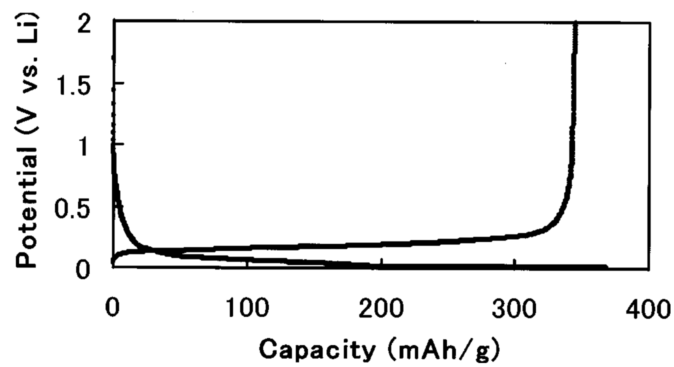

Figure 3. Charge-discharge curve of B-MCF at 100 mA/g and the first cycle.

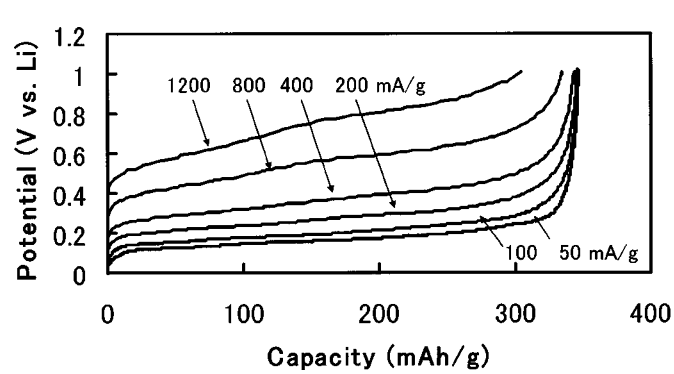

Figure 4. Discharge curves of B-MCF at various current densities.

Table II. Main specifications of laminate-type thin Li-ion battery with a thickness of 3.6 mm.

Dimensions	3.6 × 35 × 62 mm
Thickness	3.6 mm
Capacity, 0.2 C	680 mAh
Average voltage, 0.2 C	3.8 V
Weight	15 g
Charge voltage	4.2 V
1C rate current	650 mA
Cathode material	$LiCoO_2$
Anode material	B-MCF
Energy density per weight	172 Wh/kg
Energy density per volume	331 Wh/L

surface even during the high-rate charge. Therefore, we proposed that the $LiBF_4$-EC/GBL electrolyte and the B-MCF anode be used for thin Li-ion batteries using the laminated film bag.

Performance test data.—Table II shows the main specifications of the thin Li-ion battery with the 1.5 M $LiBF_4$-EC/GBL electrolyte. The thin Li-ion battery is shown in Fig. 5. The thin Li-ion battery had the energy density of 172 Wh/kg, a value higher than that of any Li-ion battery and polymer Li-ion battery currently available.[1,2,7] Figure 6 shows the discharge curves of the thin Li-ion battery at various discharge rates between 0.2 and 3C. The capacity at the 3C rate of discharge of the thin Li-ion battery was 91% of its capacity at the 0.2C rate discharge. Such a rate capability is superior to that of polymer Li-ion battery.[1,2] The discharge performance of the thin Li-ion batteries as a function of temperature between −20 and 20°C was also measured. Figure 7 shows the dependence of capacity ratio of the thin Li-ion batteries with the 1.5 M $LiBF_4$-EC/GBL, the 1 M $LiPF_6$-EC/MEC, and the 1 M $LiPF_6$-EC/DEC electrolytes on temperature. At −20°C, the capacities of the thin Li-ion battery with the 1.5 M $LiBF_4$-EC/GBL at 1C and 0.5C rate discharge were 50 and 88% of those at 20°C, respectively. The low-temperature performance of thin Li-ion battery with the 1.5 M $LiBF_4$-EC/GBL electrolyte is comparable to that with 1 M $LiPF_6$-EC/MEC electrolyte. The thin Li-ion battery with the 1.5 M $LiBF_4$-EC/GBL electrolyte provide low-temperature discharge performance superior that of polymer Li-ion batteries with a polymer gel-electrolyte.[2] The high-rate capability and the excellent low-temperature characteristics are due to the rapid deintercalation of lithium ions from the B-MCF anode and the high conductivities of the 1.5 M $LiBF_4$-EC/GBL electrolyte in the low temperature range, respectively.

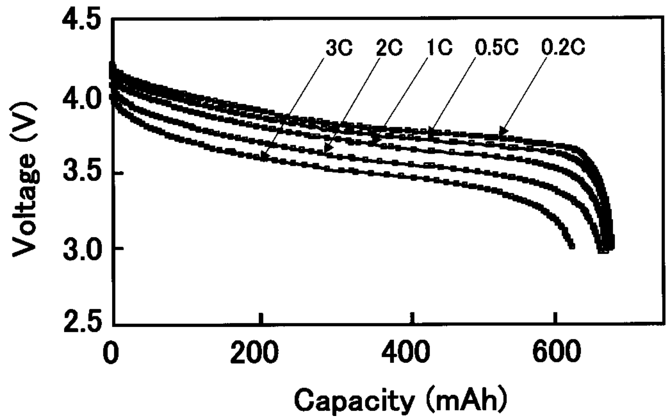

Figure 6. Discharge curves of the thin Li-ion battery at various currents.

Figure 8 shows the comparison of the cycle life of the thin Li-ion batteries using the B-MCF anode and the graphite-based (a mixture of graphite and B-MCF at 1:1 by weight) anode. The thin Li-ion battery using the B-MCF anode maintained the capacity ratio of 90% *vs.* its initial capacity after 500 cycles in a rapid charge-discharge cycling test at the 1C rate, whereas the capacity ratio of the thin Li-ion battery using the graphite-based anode was 80% after 300 cycles. It was found that lithium metal was deposited on a part of the graphite particles after 300 cycles from the result of scanning electron microscopy (SEM) observation. A passivating film with a large resistance may be formed on the graphite by a reduction of GBL solvent. We consider that lithium plating on the graphite in the EC/GBL electrolyte during the rapid intercalation is responsible for the short cycle life of thin Li-ion battery using the graphite-based anode. Thus, it has been demonstrated that the thin Li-ion battery using the B-MCF anode has good cycle life. We consider that the good cycle life of thin Li-ion batteries is attributable to the rapid lithium intercalation into B-MCF.[6,7,14]

High-temperature storage.—The thin Li-ion batteries with various electrolytes after fully charging at 4.2 V were stored for 24 h at a high temperature of 85°C to investigate swelling because the laminate film bag is expanded by the evolution of gas. The swelling of the thin Li-ion battery using the $LiBF_4$-EC/GBL electrolyte was very small, less than 0.1 mm as shown in Fig. 9. The swelling of thin Li-ion batteries using 1 M $LiPF_6$-EC/MEC and 1 M $LiPF_6$-EC/DEC electrolytes was large, more than 1 mm. The

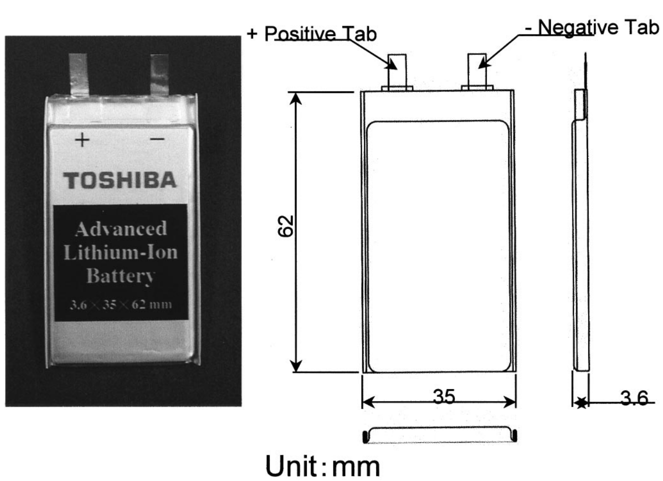

Figure 5. Photograph of laminated thin Li-ion battery with a thickness of 3.6 mm.

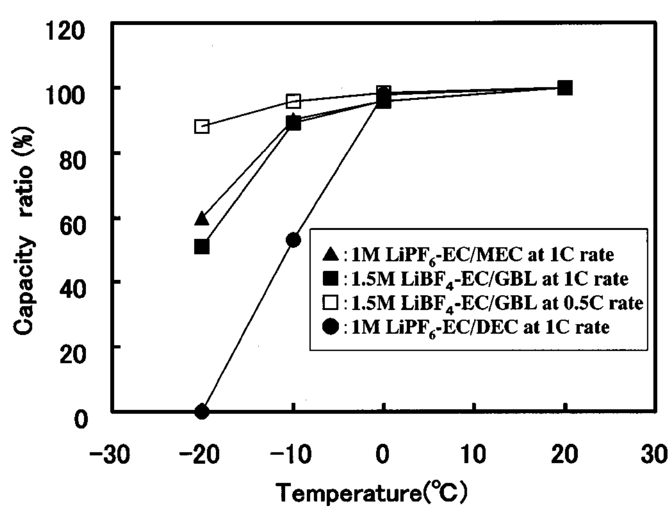

Figure 7. Dependence of capacity ratio of the thin Li-ion batteries with various electrolytes on temperature.

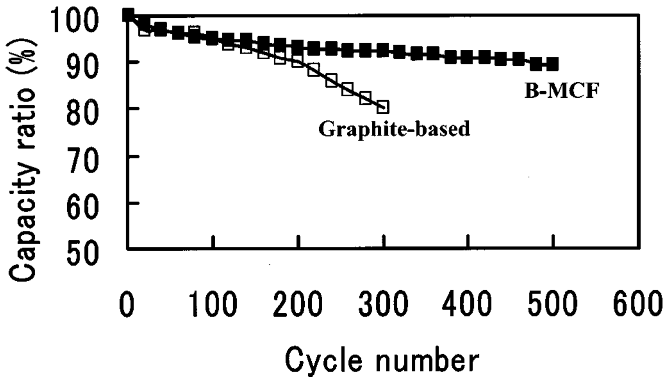

Figure 8. Discharge capacity as a function of cycle number for the thin Li-ion batteries with B-MCF and graphite-based anodes at 1C rate charge-discharge cycling.

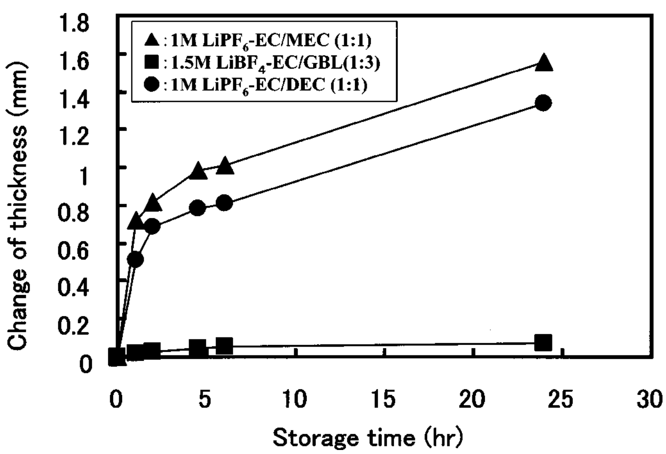

Figure 9. Change of thickness of the thin Li-ion batteries after 4.2 V fully charging with various electrolytes at 85°C storage.

$LiBF_4$-EC/GBL electrolyte is electrochemically and thermally stable under the oxidation conditions of the fully charged $LiCoO_2$ since the GBL solvent is not only thermally stable but also electrochemically stable under oxidation.[15] The $LiBF_4$-EC/GBL electrolyte in the high-temperature storage produces little evolution of gas. The volume of gas in the $LiBF_4$-EC/GBL electrolyte was so small, less than 0.1 cm^3, that we could not analyze the composition of the gas. The volumes of gas in 1 M $LiPF_6$-EC/MEC and 1 M $LiPF_6$-EC/DEC electrolytes were 2.6 and 2.2 cm^3, respectively. The DEC and MEC solvents on the fully charged $LiCoO_2$ electrode under high-temperature conditions decompose and produce large amounts of gas. The composition of the gas mainly consisted of CO_2, methane (CH_4), CO, and H_2 as shown in Table III. CO_2 was the gas whose amount was the largest. We consider that CO_2 gas is evolves by the oxidation of DEC and MEC with the full charging of $LiCoO_2$. CH_4 gas would be evolved by the reaction of the electrolyte with lithiated carbon anodes.[16] Finally, the high discharge performance, the long cycle life, and the excellent stability for high-temperature storage of thin Li-ion batteries were attributable to the high performance of B-MCF anode and the stability of the $LiBF_4$-EC/GBL electrolyte against the fully charged $LiCoO_2$ cathode material.

Table III. Gas composition in the thin Li-ion batteries using 1 M $LiPF_6$-EC/DEC and 1 M $LiPF_6$-EC/MEC electrolytes after storage for 24 h at 85°C.

Gas	1 M $LiPF_6$-EC/DEC (%)	1 M $LiPF_6$-EC/MEC (%)
CO_2	40.3	47.5
CH_4	10.7	21.8
CO	20.1	11.1
H_2	7.4	4.1
Others	21.5	15.5

Conclusions

The $LiBF_4$-EC/GBL, which is a thermally stable liquid electrolyte, and the B-MCF anode were applied to thin Li-ion batteries with a thickness of 3.6 mm using the laminated film bag. The B-MCF anode is essential for enhancing the discharge capacity and cycle life of the thin Li-ion batteries using the $LiBF_4$-EC/GBL electrolyte. The thin Li-ion batteries with high energy density of 172 Wh/kg, excellent rate performance between the 0.2 and 3C rate, a high capacity ratio at −20°C, and a long cycle life of more than 500 cycles at the 1C rate cycling have been produced. The very low swelling of the thin Li-ion batteries under high-temperature storage at 85°C was due to thermal and electrochemical stability of the $LiBF_4$-EC/GBL electrolyte against the fully charged $LiCoO_2$. The excellent performance data confirm that the $LiBF_4$-EC/GBL electrolyte and the B-MCF anode are the most promising materials for thin Li-ion batteries using the laminated film bag.

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